Bioorganic & Medicinal Chemistry Letters 22 (2012) 2401–2406

Contents lists available at SciVerse ScienceDirect

Bioorganic & Medicinal Chemistry Letters

journal homepage: www.elsevier.com/locate/bmcl

Scabronine M, a novel inhibitor of NGF-induced neurite outgrowth from PC12 cells from the fungus *Sarcodon scabrosus*

Li Liu [a], Xin-Wei Shi [a], Shi-Chun Zong [b], Jiang-Jiang Tang [a], Jin-Ming Gao [a,*]

[a] *Shaanxi Engineering Center of Bioresource Chemistry & Sustainable Utilization, College of Science, Northwest A&F University, Yangling 712100, Shaanxi, China*
[b] *Division of Medical Clinic, Xi'an University of Posts and Telecommunications, Xi'an, Shaanxi 710061, China*

ARTICLE INFO

Article history:
Received 4 December 2011
Revised 27 January 2012
Accepted 13 February 2012
Available online 20 February 2012

Keywords:
Nerve growth factor
Cyathane diterpenoids
Scabronine M
PC12 cells
Sarcodon scabrosus
Neurite outgrowth-inhibiting activity

ABSTRACT

A novel cyathane diterpenoid, designated scabronine M (**1**), was isolated from the fruiting bodies of the mushroom *Sarcodon scabrosus* together with 10 known compounds. The structure of the new compound was elucidated on the basis of extensive spectroscopic analysis including 2D-NMR. Among these compounds, only scabronine M (**1**) significantly inhibited dose-dependently NGF-induced neurite outgrowth in PC12 cells without cytotoxicity, possibly through suppressing the phosphorylation of the receptor Trk A and the extracellular signal regulated kinases (ERK). This is the first report of novel neurite outgrowth-inhibiting activity in PC12 cells of this group of diterpenoids.

© 2012 Elsevier Ltd. All rights reserved.

Neurotrophic factors (neurotrophins) such as nerve growth factor (NGF) plays an important role in differentiation, survival and maintenance of nervous system.[1] It has been shown that a decrease in availability of the NGF is considered to cause dysfunction of the nervous system, resulting in various nerve diseases including senile dementia such as Alzheimer's and Parkinson's diseases.[2] On the other hand, it was reported that up-regulation of the NGF was observed in patients with intractable temporal lobe epilepsy and Huntington's disease,[3] and glioma.[4] However, therapeutic uses of the NGF have failed because they are polypeptides, short half-lives, unable to cross the blood–brain barrier, easily metabolized by peptidases and unable to reach target tissues,[5] all made it difficult to attain the satisfied result in clinical trials. These mean the application of NGF as a medicine for treatment of neurodegenerative disorders will be uneffective and difficult. Maybe finding a small molecular weight neurotrophin agonist or antagonist is an expectant way. Therefore, both modulators, inducer and inhibitor, for neurite outgrowth can be useful to treat patients with nerve diseases. In contrast to inducers,[6] only few compounds as inhibitors of NGF-induced neurite outgrowth in PC12 cells, such as indocarbazostatin and indocarbazostatin B from a culture broth of a *Streptomyces* sp., were reported.[7]

In the preceding paper, from the mushroom *Sarcodon scabrosus* (Fr.) karst (family Hydnaceae), we reported seven cyathane diterpenoids, of which only two compounds, sarcodonins G (**1a**) and A (**1b**), exhibited neurite outgrowth-promoting activity in NGF-mediated rat pheochromocytoma PC12 cells.[6] In continuation of our ongoing search for bioactive natural products from higher fungi, we reisolated a new compound scabronine M (**1**), along with 10 known metabolites (**2–11**), from the same fungus. Among the isolated compounds, only one diterpenoid, scabronine M (**1**), had the ability to suppress NGF-induced neurite outgrowth in PC12 cells. Also, we explored the effect of scabronine M (**1**) on the receptor Trk A, and the extracellular signal regulated kinases (ERK) signal cascade. The results show that this compound suppressed the phosphorylation of Trk A receptor and the ERK. This is the first report of novel neurite outgrowth-inhibiting activity of this group of diterpenoids. In this Letter, we describe isolation, structure and biological properties of scabronine M (**1**).

An ethyl acetate soluble portion of methanol extract of *S. scabrosus* was subjected to column chromatography on silica gel and Sephadex LH-20, resulting in the isolation of a new metabolite scabronine M (**1**, 10 mg) (Fig. 1), together with 10 known metabolites (**2–11**), which was characterized by 1D, 2D NMR spectroscopy and mass spectrometry.

Scabronine M (**1**)[8] was obtained as a yellow resin. Its molecular formula was established as $C_{20}H_{28}O_4$ by HRESIMS at *m/z* 367.1660 $[M+Cl]^-$ (calcd for $C_{20}H_{28}ClO_4$, 367.1682). The IR spectrum exhibited the presence of a hydroxy (3496 cm^{-1}), a carbonyl (1703 cm^{-1}) and an α,β-unsaturated carbonyl (1686 cm^{-1}) groups. The ^{13}C NMR and DEPT spectra (Table 1) of **1** showed 20 carbon signals,

* Corresponding author. Tel./fax: +86 29 87092335.
E-mail address: jinminggao@nwsuaf.edu.cn (J.-M. Gao).

0960-894X/$ - see front matter © 2012 Elsevier Ltd. All rights reserved.
doi:10.1016/j.bmcl.2012.02.031

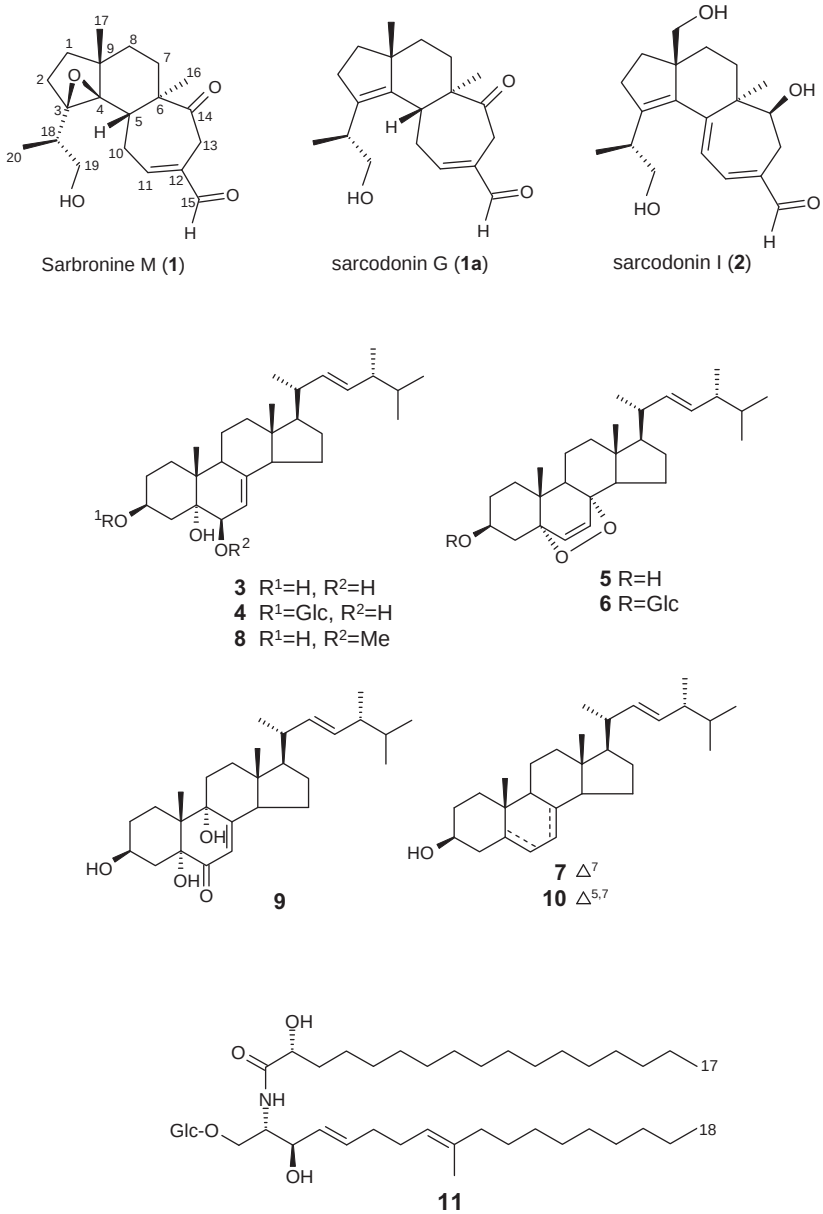

Figure 1. Structures of scabronines M (**1**), sarcodonins G (**1a**), and compounds **2**–**11**.

which were classified as three methyls, seven methylenes (one oxygenated at δ_C 67.4), four methines (one olefinic at δ_C 152.9 and one aldehyde carbon at δ_C 192.5), and six quaternary carbons including two oxygenated sp^3 quaternary carbons (δ_C 77.0 and 73.3), and one ketone carbon (δ_C 209.5).

The ^{1}H NMR spectrum (Table 1) of **1** showed signals for a secondary methyl [δ_H 1.06 (d, J = 7.1 Hz, C-20)] and two tertiary methyl groups at δ_H 1.20 (Me-16) and 1.11 (Me-17). The secondary methyl (Me-20) and two methylene protons at δ_H 3.58 (dd, J = 4.5, 10.9 Hz, H-19) and 3.88 (dd, J = 8.4, 10.9 Hz, H-19) were coupled with a methine proton at δ_H 2.18 (m, H-18), implying the presence of an isolated spin system –CH(Me)CH_2OH. Moreover, the ^{1}H and ^{13}C NMR spectra of **1** displayed an α,β-unsaturated aldehyde moiety [$\delta_{H/C}$ 6.73/152.9 (d, C-11), 136.1 (s, C-12) and $\delta_{H/C}$ 9.36/192.5 (CHO, C-15)]. This deduction was suggested by HMBC correlations of H-11 with C-13 and C-15, of H-13 and C-15, and of H-15 and C-12. The ^{1}H–^{1}H-COSY and HSQC spectra of **1** allowed the identification of four partial structures, (a) –CH(Me)CH_2O–, (b) –CH_2CH_2–, (c) –CH_2CH_2–, and (d) –$CHCH_2CH$– (Fig. 2). The connectivity of each partial structure was clarified by the HMBC spectrum as shown in Figure 2. These data allowed us to conclude that this compound may be a cyathane-type diterpenoid.[6,9] The 7° of unsaturation inherent in the molecular formula of **1**, coupled with data showing the presence of the three rings, one double bond, and two carbonyls, indicated that **1** must possess one extra ring.

Detailed analysis of the ^{1}H and ^{13}C NMR spectroscopic data revealed that **1** had a similar structure to sarcodonin G (**1a**), a previously described diterpene with cyathane skeleton identified from the same fungus.[6,9b] The significant difference between these two compounds was an epoxy group between C-3 (δ_C 77.0) and C-4 (δ_C 73.3) in **1** instead of the tetrasubstituted olefin at C-3 (δ_C 135.9) and C-4 (δ_C 141.3) in **1a**. The presence of the epoxy unit between C-3 and C-4 was supported by HMBC correlations (Fig. 2) from H-1, H-2, H-5, H-18, H-19 and H-20 to C-3 and from H-2, H-5, H-8 and H-17 to C-4.

Table 1
NMR spectral data of scabronine M (**1**) in $CDCl_3$[a]

No.	δ_H (mult; *J* in Hz)	δ_C (DEPT)	COSY	ROESY (selected)	HMBC (selected)
1	1.70 (td, 4.2, 13.8), 1.93 (td, 4.2, 13.6)	31.2 t	H-2	H-8, 16, 17	C-3, 8, 17
2	1.88 (m)	23.9 t	H-1	H-7	C-3, 4, 9
3		77.0 s			
4		73.3 s			
5	3.23 (br d, 12.5)	37.0 d	H-10	H-10, 17	C-3, 4, 11, 14
6		53.9 s			
7	1.27 (m)	33.4 t	H-8	H-1, 8, 17	C-9
8	1.34 (m), 1.46 (m)	31.1 t	H-7	H-1, 7, 17	C-1, 4
9		42.1 s			
10	2.34 (m), 2.89 (m)	27.3 t	H-5, 11	H-5, 16, 18	C-6, 12
11	6.73 (m)	152.9 d	H-10	H-5, 10, 15	C-5, 13, 15
12		136.1 s			
13	3.55 (d, 14.5), 3.74 (m)	34.5 t		H-5	C-11, 14, 15
14		209.5 s			
15	9.36 (s)	192.5 d		H-11	C-11, 12, 13
16	1.20 (s)	14.0 q		H-1, 10	C-5, 6, 8
17	1.11 (s)	19.8 q		H-18	C-1, 4, 8
18	2.18 (m)	35.0 d	H-19, 20		C-2, 3
19	3.58 (dd, 4.5, 10.9) 3.88 (dd, 8.4, 10.9)	67.4 t	H-18	H-18, 20	C-3, 18, 20
20	1.06 (d, 7.1)	13.9 q	H-18	H-18	C-3

[a] Assigned by ^{1}H NMR, ^{13}C NMR, DEPT, COSY, HSQC, NOESY, and HMBC experiments.

The relative stereochemistry of **1** was determined by NOE experiments (Fig. 2). The ROESY cross-peaks observed between H-5/H-17 and H-10 β indicated that these H-atoms are β-oriented; and the NOESY correlations of H-18/H_3-16 strongly suggested the C-18 is cofacial with the Me-16 in an α-orientation. On the other hand, the configuration of the C-3/C-4 epoxy group is assigned based on the following basis:[10] (i) examination of Drieding models indicates that the β-face of sarcodonin G (**1a**) is much less hindered than the α-face, and oxidation usually occurs from the less hindered side; (ii) the chemical shift of the C-17 methyl protons in **1** (δ_H 1.11) is downfield from the same methyl in **1a** (δ_H 1.05). These results require β-orientation of the epoxy group between C-3 and C-4. Further, this β-stereochemistry of the epoxide in **1** also was the same as that in neoallocyathin A_4, a cyathane-based diterpenoid which had been reported from a bird's nest fungi *Cyathus helenae*.[10] This is the first time to report a cyathane diterpenoid with a novel three-membered ether ring in the fungi of the Hydnaceae family.

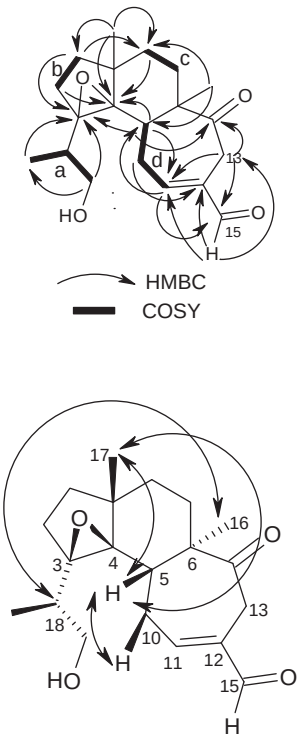

Figure 2. Important COSY, HMBC and NOE correlations of **1**.

The stereochemistry at four chiral centers C-5, C-6, C-9 and C-18 of scabronine M (**1**) was determined as 5*R**, 6*R**, 9*S** and 18*R** based on the co-occurrence of biogenetically related known metabolites, sarcodonins G (**1a**) and I (**2**), which have been obtained previously from the same mushroom.[6,9] Based on the above considerations, the structure of **1** was established as 3β,4β-epoxy-14,15-dioxo-19-hydroxycyath-11-ene, and named scabronine M.

Other known fungal metabolites, sarcodonin I (**2**),[9a] cerevisterol (**3**), tuberoside (**4**), ergosterol peroxide (**5**), 3-*O*-β-D-glucopyranosyl-5α,8α-epidioxyergosta-6,22-diene (**6**),[11] ergosta-7,22-dien-3β-ol (**7**),[12] 3β,5α,9α-trihydroxy-6β-methoxy-ergosta-7,22-dien (**8**), ergosta-7,22-dien-3β,5α,9α-trihydroxy-6-one (**9**),[13] ergosterol (**10**),[11] and a cerebroside (**11**),[14] were identified by comparing their physical and spectral data with the reported values.

The isolated metabolites (**1–11**) were evaluated for their effects on NGF-mediated neurite outgrowth in PC12 cells, and only one compound, scabronine M (**1**), was found to exhibited inhibitory activity against NGF-induced neurite outgrowth from PC12 cells. At the highest concentration (100 μM) used in this study, the compound showed no cytotoxicity by MTT assay (data not shown), suggesting that **1** is a low-toxicity agent with interesting biological activity.

In order to determine whether scabronine M (**1**) influences the NGF-mediated neuronal differentiation process in PC12 cells, we assessed cellular parameters of neuronal differentiation.[15] Scabronine M (**1**) alone failed to influence morphologically the NGF-induced neuronal differentiation in PC12 cells up to a concentration of 100 μM, this was in agreement with our previous observations.[6] As shown in Figure 3A and B, however, in the presence of an amount of NGF (20 ng/ml) that significantly induced neurite outgrowth of 21.06 ± 1.27% (Figs. 3A and B, 2a), **1** at 1 μM slightly induced neurite outgrowth of 23.22 ± 1.85% with no significant difference, while **1** significantly inhibited neurite outgrowth in a dose-dependent manner at concentrations ranging

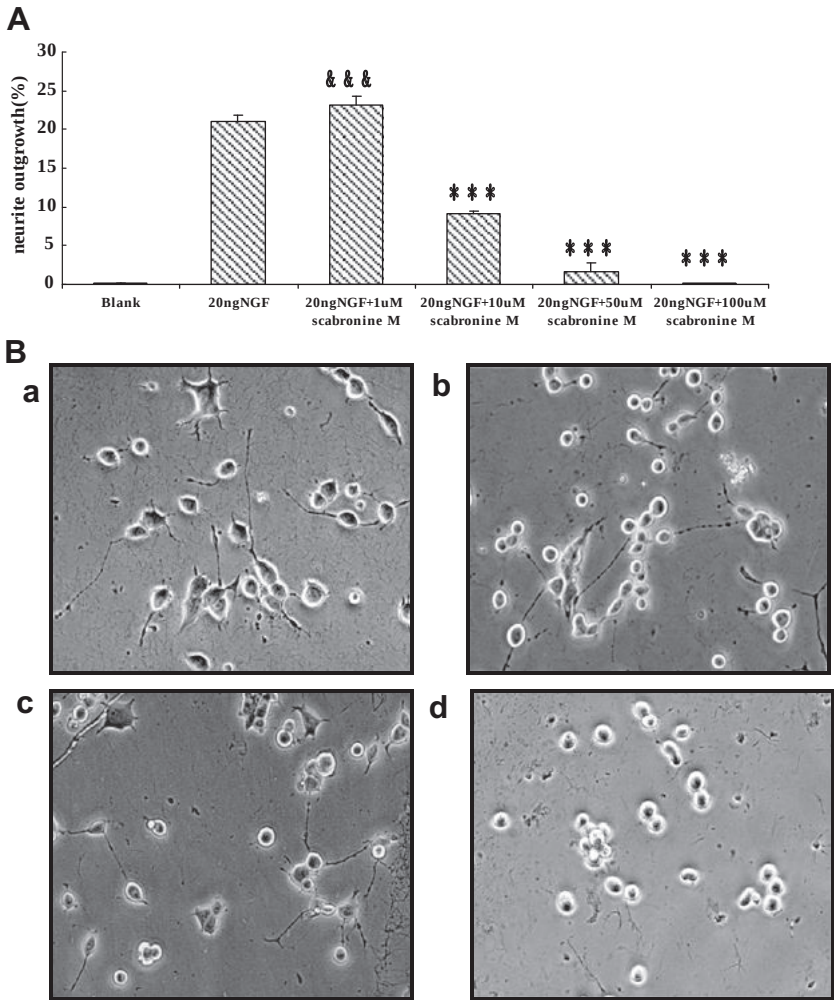

Figure 3. (A) Neurite outgrowth of PC12 cells after 24 h treatment with NGF (20 ng/ml) and scabronine M (**1**) (1–100 μM). Compound **1** at 10 μM reduced NGF-mediated neurite outgrowth after 24 h, while **1** at 1 μM had no significant effect on NGF-mediated neurite outgrowth. Blank: DMSO. (B) Representative bright-field microscopy images demonstrating the reduction of NGF (20 ng/ml)-induced neurite outgrowth by different concentrations of **1** in PC12 cells. As expected, NGF induced neurite outgrowth of the cells. (a) 20 ng/ml NGF; (b) 20 ng/ml NGF + 1 μM of **1**; (c) 20 ng/ml NGF + 10 μM of **1**; (d) 20 ng/ml NGF + 50 μM of **1**. Values were considered significant at: $^{\&\&\&}$, P <0.001 versus blank; and ***, P <0.001 versus 20 ng/ml NGF.

from 10 μM to 100 μM (Figs. 3, 2c and d). In the presence of **1** at 100 μM, the neuritogenesis reached the minimum of 0.1%, which was almost the same as blank (DMSO) (Fig. 3A). The differentiation effect (9.15 ± 0.62% at 10 μM, 1.6 ± 1.98% at 50 μM, 0.1 ± 0.05% at 100 μM) of 20 ng/ml NGF-mediated PC12 cells treated with **1** significantly decreased as the concentration of **1** became higher. Notably, **1** at both 50 μM and 100 μM almost completely inhibited the NGF-induced neurite outgrowth from PC12 cells. These results indicate that **1** potently inhibited the NGF-induced neuronal differentiation in PC12 cells rather than killing the PC12 cells.

The structural factors of **1** affecting the inhibition of NGF-induced neuritogenesis were then explored. In the present study, scabronine M (**1**), which bears an epoxy ring between C-3 and C-4, was a derivative of sarcodonin G (**1a**), showed such inhibition, while sarcodonin I (**2**) showed no inhibiting effect. Interestingly, however, we recently reported that sarcodonin G (**1a**), which contains one double bond between C-3 and C-4, can induce the NGF-mediated neuronal differentiation in PC12 cells.[6] Our findings reveal that this epoxy group may be important for activity. A subtle difference in functionalities on the cyathane structure leads to a big effect on interaction between the small molecules and PC12 cells.

The cellular mechanism of the NGF-induced neuronal differentiation has been well investigated. Trk A receptor activation leads to neurite outgrowth via phosphorylation of extracellular signal-regulated protein kinase (ERK) by mitogen-activated protein kinase (MEK) in PC12 cells.[16] The MAPKs ERK1/2 are crucially involved in cellular processes such as proliferation and differentiation[17] and ERK1/2 activation is recognized as being required for neuronal differentiation in PC12 cells.[18] Therefore, to investigate the cellular events during the suppression of NGF-induced neuritogenesis by scabronine M (**1**), phosphorylation of ERK in PC12 cells was first examined in the presence of **1** and NGF. NGF alone at a concentration (20 ng/ml) induced a phosphorylation of ERK1/2 for 15 min (Fig. 4). The relative ratio of p-ERK compared with corresponding β-actin decreased (0.26 ± 0.000 at 1 μM, 0.24 ± 0.006 at 10 μM, 0.21 ± 0.01 at 50 μM) dose-dependently, which were all lower than that of control (20 ng/ml NGF) (0.275 ± 0.005). Consequently, we demonstrate that NGF-induced ERK1/2 phosphorylation was decreased by **1**, as indicated by Western blot analysis (Fig. 4),[19] providing evidence that the NGF-stimulated activation of this signaling cascade was suppressed in PC12 cells in vitro.

On the other hand, we detected the influence of **1** on the phosphorylation of Trk A.[20] As shown in Figure 5, scabronine M (**1**) added concurrently with NGF 20 ng/ml strongly decreased Trk A phosphorylation responding to NGF. In the presence of **1** and

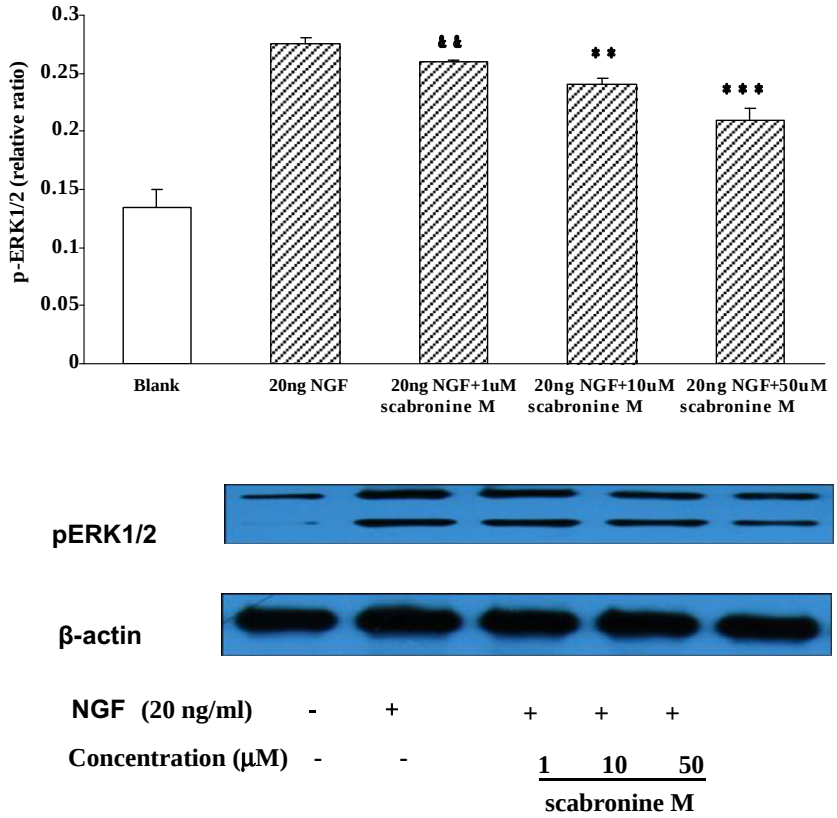

Figure 4. Western blot analysis after 15 min treatment of PC12 cells with scabronine M (**1**) (1 μM, 10 μM and 50 μM, respectively) in NGF-induced ERK1/2 phosphorylation as indicated. Each blot is a representative blot of three to six different experiments shown above densitometric analysis. Values are given as means + SEM as a relative ratio. $n = 3–6$ for all plotted values. Values were considered significant at: **, $P < 0.01$ versus control; ***, $P < 0.001$ versus control and &&, $P < 0.01$ versus control.

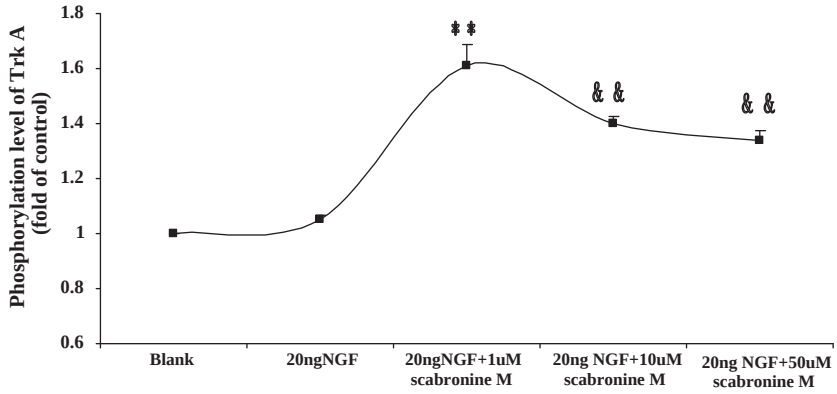

Figure 5. Phosphospecific Trk A ELISA after 15 min treatment of PC12 cells with NGF and scabronine M (**1**). Dose dependency of the suppressing effect of **1** on NGF-mediated (20 ng/ml) Trk A phosphorylation (p-Trk A; $n = 6–9$ for all plotted values). Values were considered significant at: &&, $P < 0.01$ versus 20 ng/ml NGF + 1 μM scabronine M and **, $P < 0.01$ versus 20 ng/ml NGF.

NGF, the average fold of the Trk A phosphorylation (1.61 ± 0.076 at 1 μM, 1.4 ± 0.025 at 10 μM, 1.34 ± 0.035 at 50 μM) decreased as the concentration of **1** increased. The Trk A phosphorylation of 50 μM of **1** and NGF was about 16.8% lower than that of 1 μM of **1** and NGF. These results indicate that **1** can reduce the Trk A phosphorylation in a dose-dependent manner.

In conclusion, 11 fungal metabolites were identified, including a new cyathane-based diterpenoid scabronine M (**1**). Scabronine M (**1**) was found to inhibit the NGF-induced neurite outgrowth from PC12 cell by suppressing the upstream signal p-Trk A and p-ERK. Although scabronine M (**1**) itself did not morphologically affect neurite outgrowth in PC12 cells, it inhibited potently neurite outgrowth in the presence of a small amount of NGF.

An epoxy moiety at C-3 and C-4 plays an important role in the observed biological activities. The suppression of NGF-induced neuritogenesis by scabronine M (**1**) is ascribable to decrease of the phosphorylation of ERK1/2 and Trk A in PC12 cells. Further mechanism of action of the suppression of NGF-induced neuritogenesis will be studied.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Nos. 30370019 and 30670221), the Program for New Century Excellent Talents in University (NCET-05-0852) and the Project Sponsored by the Scientific Research Foundation

for the Returned Overseas Chinese Scholars, State Education Ministry. The authors also are grateful to Ms. Joanne Magoon for the guide of activity measurements. We are also very thankful to Professor Gongshe Yang for technical assistance.

Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.bmcl.2012.02.031.

References and notes

1. (a) Huang, E. J.; Reichardt, L. F. *Annu. Rev. Neurosci.* **2001**, *24*, 677; (b) Lewin, G. R.; Barde, Y. A. *Annu. Rev. Neurosci.* **1996**, *19*, 289.
2. (a) Mufson, E. J.; Kroin, J. S.; Sendera, T. J.; Sobreviela, T. *Prog. Neurobiol.* **1999**, *57*, 451; (b) Parain, K.; Murer, M. G.; Yan, Q.; Faucheux, B.; Agid, Y.; Hirsch, E.; Raisman-Vozari, R. *Neuroreport* **1999**, *10*, 557.
3. (a) Takahashi, M.; Hayashi, S.; Kakita, A.; Wakabayashi, K.; Fukuda, M.; Kameyama, S.; Tanaka, R.; Takahashi, H.; Nawa, H. *Brain Res.* **1999**, *818*, 579; (b) Canals, J. M.; Marco, S.; Checa, N.; Michels, A.; Perez-navarro, E.; Arenas, E.; Alberch *J. Neurobiol. Dis.* **1998**, *5*, 357.
4. Meghan, C. B.; Izabela, S.; Philip, L.; George, P. T.; Luis, D. V.; Cezary, M. *Neuro-Oncology* **2008**, *10*, 965.
5. Granholm, A. C.; Albeck, D.; Backman, C.; Curtis, M.; Ebendal, T.; Friden, P.; Henry, M.; Hoffer, B.; Kordower, J.; Rose, G. M.; Soderstrom, S.; Bartus, R. T. *Rev. Neurosci.* **1998**, *9*, 31.
6. Shi, X.-W.; Liu, L.; Gao, J.-M.; Zhang, A.-L. *Eur. J. Med. Chem.* **2011**, *46*, 3112.
7. (a) Ubukata, M.; Tamehiro, N.; Matsuura, N.; Nakajima, N. *J. Antibiot.* **1999**, *52*, 921; (b) Tamehiro, N.; Matsuura, N.; Feng, Y.; Nakajima, N.; Ubukata, M. *J. Antibiot.* **2002**, *55*, 363.
8. *Data of scabronine M (**1**)*: Yellow resin ($CHCl_3$); $[\alpha]_D^{15}$ +12.7 (*c* 0.09 in $CHCl_3$); IR ν_{max} (cm^{-1}): 3496 (OH), 2962, 2932, 2873, 1703 (C=O), 1686 (C=O), 1639, 1449, 1376, 1260, 1079, 1015, 798; ^1H and ^{13}C NMR data, see Table 1. ESI-MS (negative): *m/z* 331 $[M-H]^-$; HR-ESI-MS (negative): *m/z* 367.1660 $[M+Cl]^-$ (calcd for $C_{20}H_{28}ClO_4$: 367.1682).
9. (a) Ma, B.-J.; Liu, J.-K. *J. Basic Microbiol.* **2005**, *45*, 328; (b) Shibata, H.; Tokunaga, T.; Karasawa, D.; Hirota, A.; Nakayama, M.; Nozaki, H.; Tada, T. *Agric. Biol. Chem.* **1989**, *53*, 3373; (c) Kita, T.; Takaya, Y.; Oshima, Y.; Ohta, T.; Aizawa, K.; Hirano, T.; Inakuma, T. *Tetrahedron* **1998**, *54*, 11877.
10. Ayer, W. A.; Browne, L. M.; Mercer, J. R.; Taylor, D. R.; Ward, D. E. *Can. J. Chem.* **1978**, *56*, 717.
11. Zhang, A.-L.; Xu, T.; Gao, J.-M.; Bai, M.-S.; Zhang, G.; Liu, H.-W.; Li, S.-Q.; Konishi, Y. *Heterocycles* **2009**, *78*, 1807.
12. Keller, A. C.; Maillard, M. P.; Hostettmann, K. *Phytochemistry* **1996**, *41*, 1041.
13. Kawagishi, H.; Katsumi, R.; Sazawa, T.; Mizuno, T.; Hagiwara, T.; Nakamura, T. *Phytochemistry* **1988**, *27*, 2777.
14. Gao, J. M.; Hu, L.; Dong, Z. J.; Liu, J. K. *Lipids* **2001**, *36*, 521.
15. *Neurite outgrowth bioassay*: Single PC12 cell suspensions were plated in 24-well plates at a density of 4×10^4 cells/well in serum-free culture medium and then treated with NGF at doses of 20 ng/ml and scabronine M (**1**) at 50, 10 and 1 μM, respectively. After 24 h incubation, the percentage of cells showing neurite outgrowth was determined by light microscopy. Cells with one or more neurites whose lengths were at least 1.5-fold the diameter of the cell body were scored as positive. Neurite outgrowth was determined from at least three different regions of interest in three independent experiments.
16. Traverse, S.; Gomez, N.; Paterson, H.; Marshall, C.; Cohen, P. *Biochem. J.* **1992**, *288*, 351.
17. (a) Levi-Montalcini, R. *Science* **1987**, *237*, 1154; (b) Rosa, R.; Garcia, A. A.; Braschi, C.; Capsoni, S.; Maffei, L.; Berardi, N.; Cattaneo, A. *Proc. Natl. Acad. Sci. U.S.A.* **2005**, *102*, 3811; (c) Weikert, T.; Rathjen, F. G.; Layer, P. G. *J. Neurochem.* **1994**, *62*, 1570.
18. (a) Hempstead, B. L.; Martin-Zanca, D.; Kaplan, D. R.; Parada, L. F.; Chao, M. V. *Nature* **1991**, *350*, 678; (b) Wolf, D. E.; McKinnon, C. A.; Daou, M. C. *J. Biochem.* **1998**, *37*, 3178.
19. *Western blot analysis*: Cells was performed as described previously. Briefly, PC12 cells were washed twice in ice-cold PBS and lysed. Total proteins were isolated by a commercial kit (Keygen Biotechnology, Nanjing, China) according to the manufacture's instruction. Cell lysates were matched from protein content, and equal amounts of protein were loaded on each lane, separated by SDS–PAGE, and transferred onto PVDF membrane. After blocking, membranes were immunostained with rabbit anti-phospho-ERK1/2 antibody (1:400; Santa Cruz Biotechnology, CA, USA), and rabbit anti-β-actin antibody (1:5000; Keygen Biotechnology, Nanjing, China) and shaked overnight at 4 °C. After washing, the PVDF membranes were incubated with 1:5000 dilution of the goat anti-rabbit horseradish-peroxidase coupled secondary antibody (Keygen Biotechnology, Nanjing, China) for 1 h at room temperature, and then used the PBST to rinse the membranes. Added developer to PVDF membranes, placed 1 min at room temperature and wrapped them with cling film. The membrane protein were attached to the X-ray film quickly and exposed in dark condition. Adjusting the exposure time, until the best appears.
20. Quantification of Trk A phosphorylation by ELISA: Cells were treated according to aforementioned procedures at density of 1×10^6 cells/ml, washed twice in ice-cold PBS and lysed by freeze thawing. Cell lysates was detected by ELx800 Immunoassay System (BIO-TEK, USA). The absorbance was measured at 450 nm.